

Program : Diploma in Tool and Die Engineering	
Course Code : 3112	Course Title: Production Engineering II
Semester : 3	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To introduce the basic production processes and technologies of relevance to the manufacturing industry and related sectors
- To impart knowledge on detailed machining aspects so as to select, operate and control the appropriate machining method and surface finishing processes for specific applications

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic Tooling		Production engineering -I	2
Engineering science		Engineering workshop practice	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the functions and machining operations on lathe and drilling machine	15	Applying
CO2	Describe the functions and machining operations of Planer, Shaper and Slotter	15	Understanding
CO3	Interpret various operations of milling machines and gear generating processes	13	Applying
CO4	Illustrate abrasive processes and various types of operation on grinding machines	15	Applying
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	2						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the functions and machining operations on lathe and drilling machine		
M1.01	To illustrate basics parts and accessories of Lathe and its functions	4	Applying
M1.02	To demonstrate different Lathe operations	4	Applying
M1.03	To describe the working of Drilling machine	3	Understanding
M1.04	To describe different drilling operation	4	Understanding
Contents: Lathe : Types of lathes; Basic parts and their functions; Specifications ; Work holding and tool holding device , Operations - Turning, parting off, Knurling, facing, Boring, drilling, threading, step turning, taper turning. Drilling Machines: Drills - Flat drills - Twist drills - Types of drilling machines - Bench type - Floor type - Radial type - Gang drill - Multi - spindle type - Principle of operation in drilling - Speeds and feeds for various materials - drilling holes - methods of holding drill bit - drill chucks - socket and sleeve - drilling - operation - reaming - counter sinking - counter spot facing - tapping - deep hole drilling.			
CO2	Describe the functions and machining operations of Planer, Shaper and Slotter		
M2.01	To describe the working of shaper	4	Understanding
M2.02	To describe the working of Planer	4	Understanding
M2.03	To explain the working of slotter	3	Understanding
M2.04	To describe quick return mechanism-crank and slotted link-feed mechanism used in shaper, planer, and slotter	4	Understanding
	Series Test - I	1	

Contents:

Shaper: Types of shapers-specifications-standard-plain-universal principles of operations-drives-quick return mechanism-crank and slotted link-feed mechanism-work holding devices-special fixture-various shaper operations.

Planer: Types of planers - description of double housing planer specifications - principles of operation- drives -quick return mechanism-feed mechanism- work holding devices and special fixtures-types of tools and various planer operation

Slotter: Types of slotters-specifications-method of Operation-Whitworth quick return mechanism- feed mechanism-work holding devices-types of tools

CO3	Interpret various operations of milling machines and gear generating processes		
M3.01	To demonstrate the functions and operations of milling machines	4	Applying
M3.02	To describe the work, tool holding devices and milling cutters	3	Understanding
M3.03	To demonstrate milling process and milling operations	3	Applying
M3.04	To explain Gear Generating Processes	3	Understanding

Contents:

Milling Machines: Types-column and knee type-plain- universal milling machine-vertical milling machine-specification of milling machines principles of operation-work and tool holding devices- arbor-stub arbor spring collet-adapter-milling cutters-cylindrical milling cutter-slitting cuttersidemilling cutter-angle milling cutter-T-slot milling cutter-woodruff milling cutter-fly cutter-nomenclature of cylindricalmilling cutter-milling processconventional milling-climb milling-milling operations-straddle milling-gang milling-vertical milling attachment.

Gear Generating Processes: Gear Milling-Gear hobbing--Gear finishing processes-Burnishing-Shaving-Grinding and Lapping; Gear Materials-Cast iron, Steel, Alloy steels, Brass, Bronze, Aluminum and Nylon

CO4	Illustrate abrasive processes and various types of operation on grinding machines		
M4.01	To describe the aspects of abrasive processes	3	Understanding
M4.02	To demonstrate various grinding operations.	5	Applying
M4.03	To Illustrate the selection of grinding wheel	3	Applying
M4.04	Explain the micro finishing operations	4	Understanding
	Series Test – II	1	

Contents:

Abrasive Processes : Types and classification - specifications - rough grinding - pedestal grinders - portable grinders - belt grinders - precision grinding cylindrical grinder - centerless grinders - surface grinder - tool and cutter grinder - planetary grinders - principles of operations - grinding wheels abrasives - natural and artificial diamond wheels - types of bonds - grit, grade and structure of wheels - wheel shapes and sizes - standard marking systems of grinding wheels - selection of grinding wheel - mounting of grinding wheels - Dressing and Truing of wheels - Balancing of grinding wheels.

Finishing: micro finishing operations - honing, lapping, superfinishing.

Text/ Reference:

T/R	Book Title/Author
T1	Introduction of Basic Manufacturing Processes and Workshop Technology, Rajendersingh, New age International (P) Ltd. New Delhi- 110002, 2006
R1	Elements of Workshop Technology Volume I & II, HajraChowdry& Bhatt Acharaya, Media Promoters, 11th Edition, 2007
R2	Manufacturing Process Begeman, Tata McGraw Hill, New Delhi.
R3	Production Technology –H.M.T,Khanna Publishers.
R4	A Text Book of Production engineering – P.C. Sharma, S.Chand& Co.

Online Resources:

Sl.No	Website Link
1	http://www.mit.edu/
2	https://nptel.ac.in/